

Dynamic ensemble selection for multi-class imbalanced datasets

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ABSTRACT

Many real-world classification tasks suffer from the class imbalanced problem, in which some classes are highly underrepresented as compared to other classes. In this paper, we focus on multi-class imbalance problems which are considerably more difficult to address than two-class imbalanced problems. On this account, we develop a novel and effective procedure, called dynamic ensemble selection for multi-class imbalanced datasets (DES-MI), in which the competence of the candidate classifiers are assessed with weighted instances in the neighborhood. The proposed DES-MI consists of two key components: the generation of balanced training datasets and the selection of appropriate classifiers. To do so, we develop a preprocessing procedure to balance the training dataset which relies on random balance. To select the most appropriate classifiers in the scenario of multi-class imbalance problems, we propose a weighting mechanism to highlight the competence of classifiers that are more powerful in classifying examples in the region of underrepresented competence. We develop a thorough experimental study in order to verify the benefits of DES-MI in handling multi-class imbalanced datasets. The obtained results, supported by the proper statistical analysis, indicate that DES-MI is able to improve the classification performance for multi-class imbalanced datasets.

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1. Introduction

In the research community of machine learning, the class imbalance learning problem is considered to be one of the most important challenges [22]. Imbalanced data refers to such a dataset in which several classes are underrepresented (minority classes) when compared with others (majority classes) [37]. This skewed distribution makes many classic classification algorithms, all of which designed with the premise of a balanced training dataset, such as Bayes classifier, decision tree, k -nearest neighbor (k NN), artificial neural network (ANN), support vector machine (SVM), less effective [47]. On the other hand, many real-world classification tasks involve the class imbalance learning problems, for example, social media analysis [25], action recognition [24], water quality prediction [26], and text categorization [43].

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In fact, many efforts have been devoted to imbalanced learning problems in the research community, such as resampling methods, cost sensitive approaches, ensemble learning algorithms, kernel-based methods and active learning methods [15]. However, most of them so far have been focused on two-class imbalanced datasets. Obviously, multi-class imbalance learning problems are much more difficult to address than the binary scenario, since the decision boundary involves distinguishing between more classes. Unfortunately, directly applying the methods proposed for dealing with two-class imbalanced problems to the multi-class problems may be invalid [9].

Three types of difficulties related to a multi-class imbalanced dataset exist: one majority and many minority classes, one minority and many majority classes, and many minority and many majority classes [31]. In addition, the skewed distribution of instances among classes is not the only source of difficulties for classification algorithms to deal with multi-class imbalanced datasets. Difficulties embedded in the structure of data are also always present, such as class overlapping, small disjuncts (minority class can consist of several sub-concepts) and small sample size (lack of representative minority instances).

As explained above, in multi-class imbalance learning tasks, different characteristics are represented in different regions, which are associated with different classification difficulties. Therefore, it is reasonable for us to adopt a dynamic scheme to select different classifiers for different query examples. Dynamic classifier selection (DCS) and dynamic ensemble selection (DES) are the most famous techniques based on dynamic selection [49]. The former tends to select the most appropriate single classifier for the query instance, while the latter aims to dynamically acquire a classifier system consisting of several competent classifiers. The latest research results [4] indicate that DES usually outperforms the DCS method, since the former approach distributes the risk of this over-generalization by selecting a group of classifiers instead of one individual classifier for a query example.

Motivated by the above analysis, a simple and yet effective method named DES-MI, which stands for dynamic ensemble selection for multi-class imbalanced datasets, is proposed in this paper to deal with multi-class imbalanced datasets, paying attention to the following tasks:

- The diversity of classifiers in the candidate pool is achieved based on the random balance framework, which hybridly uses the techniques of random under-sampling (RUS), random over-sampling (ROS) and synthetic minority oversampling technique (SMOTE) [5].
- Then, the competence of candidate classifiers is evaluated by using the weighted instances in the neighborhood surrounding the query example x_t . That is, we consider higher competence for the classifier that is more powerful in classifying minority classes within the local region.
- Finally, every selected classifier submits a vote on the query example x_t . The votes received by each class are counted and the class with the largest number of votes is considered as the final output class.

In order to show the validity of our approach, we carry out a thorough experimental study on 20 multi-class imbalanced datasets selected from the KEEL dataset repository [36]. We measure the performance of the methods based on both macro average arithmetic (MAvA) [32] and mean F-measure (MFM) [12] metric. The effectiveness of dynamic selection in the scenario of multi-class imbalanced datasets is analyzed and DES-MI is compared with the state-of-the-art methods which are also developed for solving the multi-class imbalanced problem. The significance of the results is studied by the use of the proper statistical tests as suggested in [7].

The main contributions of this paper can be summarized as follows:

- In this paper, we focus on multi-class imbalance learning tasks, which are considerably more complicated than the binary scenario. A comprehensive analysis of difficulties in multi-class imbalanced datasets is presented, which provides a deep insight into the nature of multi-class imbalanced datasets.
- We propose a novel method to deal with multi-class imbalance learning problems by using a dynamic scheme, in which a group of appropriate classifiers are selected for each query example.
- Since classic DES approaches are developed for balanced datasets, we in this paper present a new DES model for imbalanced multi-class datasets, which generates the classifier pool by using a novel data preprocessing method. Then, it associates the voting weights of instances in the neighborhood of the query example which are adaptively adjusted according to the data distribution.

The remainder of this paper is organized as follows. In Section 2, we first describe the multi-class imbalance learning problem, including the notations used throughout this paper and solutions for multi-class imbalance learning tasks. Next, in Section 3 we review related works on DES. Then, Section 4 presents the proposed method, DES-MI. Section 5 describes the experimental framework, whereas Section 6 shows the results and discussion. Finally, the conclusions are given in Section 7.

2. Multi-class imbalance learning problem: notations and solutions throughout the literature

In order to clearly present our findings, we first establish the following notations used in this paper:

- $D = \{x_i, y_i\}_{i=1}^n$ is a dataset containing n examples, where $y_i \in \{\omega_1, \dots, \omega_m\}$ is the corresponding class label for the sample x_i and m is the number of classes.
- x_t is a test example with an unknown class label.

- T is the number of base classifiers in the candidate classifier pool.
- $\mathcal{L}$ is the base learner.
- $\mathcal{H}$ is the candidate classifier pool.
- EoC_t^* is an ensemble of selected classifiers for each example x_t in D_{te} .
- Ψ is the region of competence.
- w_i is the voting weights for each x_i in Ψ .
- $I(\cdot)$ is the indicator function, where $I(\cdot) = 0$ or 1 , according to whether the statement I is false or true.
- $C(h_j|x_t)$ is the competence of classifier h_j for the classification of x_t .
- $H(x_t)$ is the obtained output for a tested sample x_t .
- $\mathcal{O}(\cdot)$ is the time complexity.

Considering a given dataset D with n examples (i.e. $|D| = n$), we formally define $D = D_1 \cup D_2 \cup \dots \cup D_m$, where $D_i = \{x_j, y_j\}$, $j = 1, \dots, n_i$, and $x_j \in X$ is the characteristic vector of each example, $y_j \in Y = \{\omega_1, \dots, \omega_m\}$ is the class label associated with the example x_j , m is the number of classes, and n_i is the number of examples with the class ω_i . When the classification problem involves more than two classes and several classes (minority classes) are underrepresented with respect to the other classes (majority classes), i.e., if $|D_i| \ll |D_j|$, $i \neq j$ and $i, j = 1, \dots, m$, the classification task on D is called multi-class imbalance problem.

When we refer to multi-class imbalance classification problems, the canonical solutions may not be effective or achieve a lower performance, since it is easy to lose performance in one class while trying to gain it in others [9]. Therefore, it is necessary to develop a feasible methodology to handle these problems. Recently, some studies have been devoted to handling this task that can be roughly categorized into two groups:

2.1. Straightforward solutions

In this framework, the solutions focus on modifying existing algorithms to strengthen their ability to learn from multi-class imbalanced datasets. Among them, AdaBoost.NC with random oversampling [37] is a representative method which combines negative correlation learning and boosting to penalize classification errors and encourage ensemble diversity. Static-SMOTE and dynamic-SMOTE, which combine the over-sampling techniques, were proposed to address the multi-class imbalance problems [11]. Static-SMOTE applies the resampling procedure for m steps in the preprocessing stage, where m is the number of classes. In each iteration, the resampling procedure first selects the class with minimum size, and then adds the same number of instances as present in the original dataset by applying the SMOTE algorithm. Dynamic-SMOTE first changes the dataset before the algorithm performs taking into account the number of patterns per class, and then the size of instances with the minimum sensitivity increases iteratively in different generations according to the usage of the SMOTE algorithm. A hybrid system was developed by the usage of feature selection based on the binary particle swarm optimization (BPSO) algorithm, combining AdaBoost and k -NN, in which a normalized AUC was used to evaluate the fitness of the individual [16]. An adapted ensemble learning system [46] was designed to handle the multi-class imbalance learning problems, in which four issues were considered when building a multiple classifier system, including the choice of base classifiers, ensemble rules, pruning classifiers and feature space. The Mahalanobis distance-based over-sampling (MDO) technique was developed in [2], which over-samples the minority class by considering each candidate minority class sample and generating a new synthetic instance that has equal Mahalanobis distance from the considered class mean to the candidate sample, while in [1] MDO was embedded into the boosting procedures. In [31], four different types of examples in multi-class datasets were detected: safe, borderline, rare and outliers, whose combinations guide the introduction of synthetic examples. With respect to cost-sensitive learning, over-sampling, under-sampling and threshold-moving cost-sensitive neural networks were developed in [50] via balancing the cost of misclassification. In addition, a cost-sensitive boosting algorithm derived from AdaC2 was developed to improve the classification performance for imbalanced datasets involving multiple classes [34]. Furthermore, a new method named boosting adapted for cost-matrix (BAAdaCost) [10] was presented by introducing multi-class cost-sensitive margin and a procedure to estimate the cost matrix from the 0/1-loss confusion matrix.

2.2. Decomposition strategy

These techniques aim to deal with the multi-class imbalance problems by using a decomposition strategy to divide the more complicated original problem into several easier-to-solve binary class sub-problems [13]. Therefore, the massive approaches proposed for coping with binary class imbalanced problems can be used to handle the multi-class imbalanced problems. A novel decision tree for multi-class imbalance problems was built by using a decomposition scheme and Hellinger distance as the splitting criterion [19]. In [9], a thorough experimental analysis of the use of the combination of binarization techniques, i.e. one versus one and one versus all, with both preprocessing of instance and cost-sensitive learning with respect to several ad hoc approaches, was carried out. The results have shown the good behavior achieved by the synergy between pairwise learning and oversampling/cost-sensitive learning. Recently, in [48] multi-class imbalanced data was handled by using ensemble learning techniques for each sub-problem derived from decomposition strategy. Besides, a method named near-Bayesian support vector machines (NBSVMs) [6] was proposed to address the multi-class imbalanced

dataset by the usage of combination of pairwise modified SVMs with boundary shift as well as asymmetric regularization costs using one-vs-all technique.

However, after careful analysis we can easily find that all the previous research belongs to the static version. In fact, in the multi-class imbalance learning problem, different query examples usually have different classification difficulties. Therefore, if we can employ the dynamic strategy to select the most appropriate classifiers for each query example, the classification performance may be better than the simple static strategy. This considered, it is reasonable for us to use the DES framework, which is described in the next section, for the multi-class imbalanced datasets.

3. Dynamic ensemble selection

The multiple classifier systems (MCSs), which are composed of a pool of base classifiers, are shown to outperform its members across a wide range of classification problems [20]. The construction of a MCS can be divided into three phases: (1) generation of a pool of base classifiers, (2) selection of a single classifier or a subset of classifiers, (3) integration of the selected classifiers. We refer to the subset of classifiers as the ensemble of classifiers (EoC), which aims to find an appropriate ensemble with the best classifiers. The ensemble selection phase can be either static or dynamic. In static ensemble selection, such as in boosting methods and in decision forests, the ensemble is selected according to the selection criteria during the training phase, and is used to classify all query examples in the generalization phase. In contrast, the DES family of techniques attempts to find an ensemble of classifiers C' that has the most competent classifiers from a pool of classifiers C , where $C' \subset C$, for a specific query example x_i . As different classifiers usually have different classification capabilities in classifying different examples, the fact that DES-based techniques select an appropriate group of classifiers for each query example makes them a robust approach for handling a number of real-world applications.

There are two main ensemble types which generate the candidate classifiers: homogeneous and heterogeneous. The former consists of classifiers trained by using the same classification algorithm on different datasets, whereas the latter is obtained by the usage of different classification algorithms on the same training dataset. Considering the dynamic selection procedure, the most important component of DES-based techniques is the selection criterion that is used to measure the competence of classifiers. The most common approach used to estimate the competence of classifiers is to adopt the accuracy of candidate classifiers in a small region of the feature space surrounding the query example needed to be classified. The K-nearest-oracles (KNORA) [21] family of techniques uses the oracle information to measure the competence of the classifier. Wołoszynski et al. proposed several approaches that use the probabilistic model to evaluate the classifier competence such as DES-Logarithmic-function (DES-LF), DES-Exponential-Function (DES-EF), DES-Piecewise-Linear-Function (DES-PLF) [40], DES-competence based on label outputs and weighted majority voting (DES-CV) and DES-competence based continuous-value outputs and weighted class supports (DES-CS) [41]. Furthermore, measures of competence based on random classification (MCR) were developed such as DES-Performance (DES-P) and DES-Kullback-Leibler (DES-KL) [42]. Some studies have shown that it is important to make a tradeoff between accuracy and diversity when constructing a MCS. Santana et al. [33] proposed to obtain an ensemble for each query example by selecting the most diverse classifiers from the pool of classifiers that are the most accurate in the local region. A probabilistic model based on randomized reference classifier (RRC) was proposed to calculate accuracy and diversity [28]. In addition, there are some proposals developed to estimate the competence of a group of classifiers rather than that of an individual classifier. Dos Santos et al. [8] proposed a dynamic overproduce-and-choose strategy, which dynamically selects an ensemble from a population produced by an evolutionary algorithm. Recently, Mousavi and Eftekhari [30] developed a hybrid approach that employs the DES-P procedure to perform on the candidate classifiers which are selected by an optimized static ensemble selection (SES).

Although there has been a lot of research devoted to building a DES classifier system, most of it does not take into consideration a skewed data distribution. Xiao et al. [45] developed a DES classifier using cost-sensitive selection criteria for binary class imbalanced datasets. To the best of our knowledge, there are no efforts focusing on the construction of a DES system for the multi-class imbalance problems. Motivated by the good performance of DES methods in handling balanced datasets, we aim to develop a DES-based system in the next section for the multi-class imbalanced datasets.

4. The proposed method: DES-MI

In this section, we first present some preliminary concepts of the difficulties in multi-class imbalanced datasets to elicit our motivation as to why we consider using DES-based frameworks to deal with multi-class imbalance learning problems. Then, we present the details of the proposed method.

4.1. Preliminary and hypothesis

When we refer to multi-class imbalance classification problems, the canonical solutions with good behavior in two-class imbalanced datasets may not be effective or achieve a lower performance, since it is easy to lose performance in one class while trying to gain it in others [9]. The main challenges in multi-class imbalance learning problems include ambiguity, rare instances (lack of representative data), small disjuncts, noise instances and multi-class overlapping.

The first issue in multi-class imbalance learning problems is the ambiguity that the relationship among the classes is no longer clear. In two-class imbalanced datasets, the class considered the negative class severely outrepresents the other,

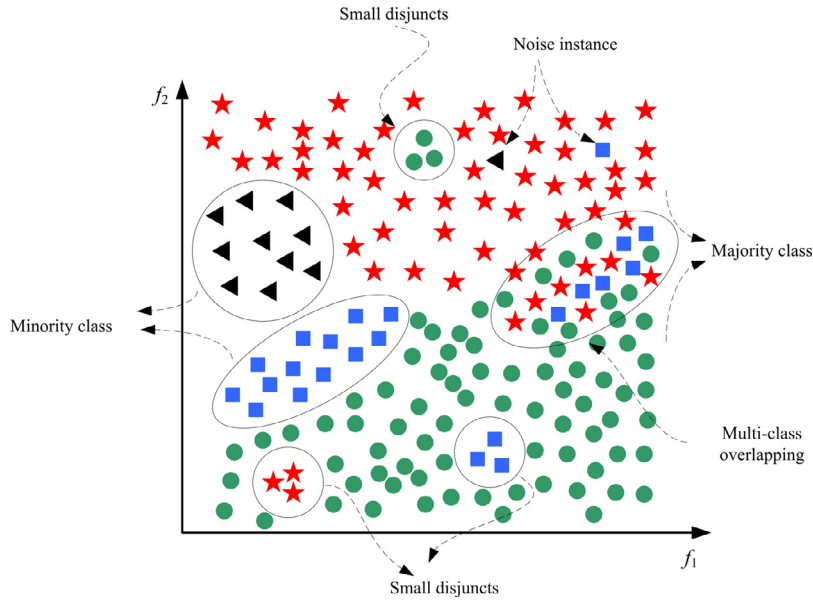


Fig. 1. Example of difficulties in a multi-class imbalanced dataset.

which is considered the positive class. Therefore, in the case of binary scenario, it is imperative for us to distinguish the positive class from the negative class. However, the relationship among classes in multi-class imbalance problems are considerably more complex than that in the binary scenario, since the former involved in more than two classes needed to be distinguished. In order to clearly illustrate the multi-class imbalance scenario, suppose we have an imbalanced dataset with four different classes as shown in Fig. 1, where we represent an artificial 2-dimensional dataset, in which the stars, circles, squares and triangles represent examples with four different classes. In general, the stars and circles could be considered the majority classes, while the others are negative classes, since the former examples outnumber the latter. However, for example, the square class is considered the minority one when compared with the star and circle classes, while it is considered the majority class when compared with the triangle class.

Besides the intricate between-class imbalance in multi-class learning tasks, another issue in this scenario is that rare instances, i.e., the examples with minority classes, are very limited. In this situation, the lack of representative data with respect to the target concept makes the learning task difficult regardless of the between-class imbalance [17]. Furthermore, in the scenario of multi-class imbalanced datasets, the number of the target concepts suffered from rare instances might involve multiple classes, which greatly depreciates classification performance.

It is worth noting the differences between small disjuncts and noisy examples. It is desirable for an induced classifier to cover the small disjuncts, while noisy examples should be ignored since they negatively affect the structure and performance of a classifier. However, it is difficult to learn from noisy and imbalanced data because noisy examples may cause a number of erroneous small disjuncts to be induced [38]. For example, in Fig. 1, there are three small disjuncts and two noisy examples. Suppose a classifier generates disjuncts for each of the two noisy examples, then these disjuncts would be illegitimate with respect to the other disjuncts which are legitimately formed from severely underrepresented subconcepts.

The last issue in the multi-class imbalanced dataset we would like to discuss is the overlapping problem. The overlapping problem occurs when a region of data space contains a similar number of training examples for each class. This makes it difficult or impossible to distinguish between the classes, since the prior probabilities of each class in the overlapping region are near equal. Some studies [14] have shown that the overlap might be a much more serious issue than the problem of classes represented by a number of many different examples. Especially in the scenario of multi-class imbalanced datasets, the overlapping problem becomes more severe as the number of classes increases, which leads to an overlap with more than two classes. For example, in Fig. 1 there is a multi-class overlapping, in which different classes are embedded, i.e. the star, circle and square classes.

As analyzed above, different characteristics are represented in different regions associated with different classification difficulties in multi-class imbalanced datasets. Therefore, it is reasonable for us to assume that the classification ability of the classifier system might be better if it is fit to different classifiers rather than to a static version. This has led us to believe that using DES-based techniques might obtain better results when handling multi-class imbalanced datasets. As far as we know, no effort has been made to explore the appropriateness of the properties of DES-based techniques for multi-class imbalance problems. Therefore, we attempt to deal with multi-class imbalanced datasets using a methodology based on a DES framework, which is described in the following section.

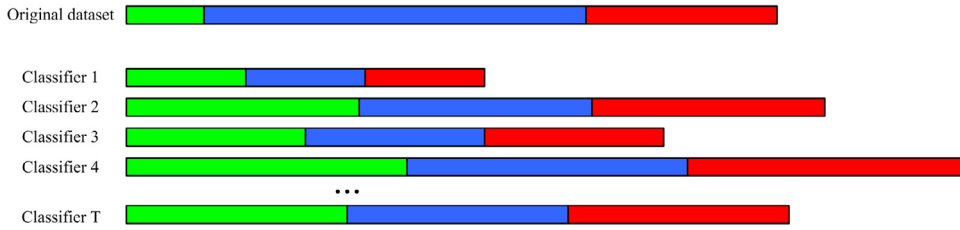


Fig. 2. Example of the proposed preprocessing to generate multi-class balanced datasets for obtaining the candidate ensemble.

4.2. Description of DES-MI

In this subsection, we present our proposed approach DES-MI for multi-class imbalance learning problems based on dynamic selection of classifiers, which consists of two key components as follows:

- *Generation of the candidate classifiers.* As stated above, the inherent mechanism behind DES-based methods is known as diversified classifiers performing differently in different regions. In this study, we consider employing the homogeneous ensemble to generate the candidate classifier pool. This considered, we propose a novel preprocessing technique which relies on random balance to restore the balance of the class proportions. In this manner, the diversities among the candidate classifiers are obtained with the selected classification algorithm in the different training datasets.
- *Dynamic selection of the most appropriate ensemble.* In classic DES methods, the distribution of instances within the region of competence is not considered, and it makes the final decision towards the majority classes. In order to extend the DES approach into a multi-class imbalanced scenario, we develop a novel weighting method to outstand the competence of a candidate classifier with more power in classifying the minority classes. That is, the instances belonging to the minority classes within the neighborhood of a query example have greater weights when evaluating the level of competence of a classifier in the candidate classifier pool.

4.2.1. Generation of candidate classifiers

Classic preprocessing techniques can have inevitable drawbacks. RUS might neglect some potentially useful information contained in the majority class examples, while ROS can increase the training time and cause the problem of overfitting. SMOTE can be seen as a variant of the oversampling technique which might introduce noise in the synthetic minority class examples.

While the homogeneous ensemble is used to generate the candidate classifiers pool, preprocessing techniques are important to be able to obtain the diversified training dataset for each base learner. Considering the scenario of multi-class imbalanced datasets that this paper focuses on, we propose a novel preprocessing procedure which combines the undersampling and oversampling techniques. The motivation of the proposed preprocessing is to generate adequately diverse training datasets. In this way, the adequate diversity among candidate classifiers can be obtained.

An example of the proposed preprocessing can be seen in Fig. 2. Given a dataset, a balanced dataset of different sizes is obtained for each member of ensemble. In this example, the initial proportion among classes appears on the top. Classifier 1, 2, ..., T are trained using variants of the initial dataset where the class distribution is balanced. It is interesting to note that the size of the variant varies randomly.

The procedure of generating a balanced dataset used for training each member of the candidate ensemble is described in the pseudocode in Algorithm 1. Suppose that a initial dataset $D = \{x_i, y_i\}_{i=1}^n$ containing n examples is given, where x_i is an input feature vector and $y_i \in \{\omega_1, \omega_2, \dots, \omega_m\}$ represents the corresponding class label of x_i . The proposed preprocessing method attempts to generate balanced training datasets with adequate diversities among them. With this objective, the key component is to randomly set the new size for each class, which is equal to a random integer value between 1 and the size of the largest class (lines 6–7). If the size of the investigated class is greater than the random integer, then a RUS procedure without replacement is used to reduce the size of the current class (lines 11–14), otherwise a ROS procedure is used to increase the size of the current class. (lines 16–17). Further, in order to obtain a balanced training dataset, we employ the procedure as used in the SMOTE method, which produces synthetic examples for the class whose size is smaller than the largest one. Concretely, for each minority example, its k (which is set to as used in SMOTE) nearest neighbors of the same class are selected and new synthetic examples are interpolated between the minority example and its random selected nearest neighbors (line 20). Afterwards, the selected base learner is trained using the generated balanced dataset.

4.2.2. Dynamic selection with the weighting mechanism

The aim of dynamic selection is to select the most appropriate ensemble from the candidate classifier pool for each query example, x_r . This considered, the most important component is how to assess the competence of candidate classifiers. Although a large number of methods have been proposed to measure the classifier's competence, all of them were designed with the premise of a balanced scenario.

Algorithm 1 Pseudocode for the generation of candidate classifiers.

```

1: Input: Initial dataset  $D = \{x_i, y_i\}_{i=1}^n$  and  $y_i \in \{\omega_1, \dots, \omega_m\}$ , base learner  $\mathcal{L}$ , the number of base classifiers  $T$ 
2: Output: Candidate classifier pool  $\mathcal{H}$ 
3:  $\mathcal{H} \leftarrow \emptyset$ 
4: for  $t = 1$  to  $T$  do
5:    $D_{new} \leftarrow \emptyset$ 
6:    $MajoritySize \leftarrow$  Obtain the size of the largest class
7:    $NewMajoritySize \leftarrow$  Random integer value between 1 and  $MajoritySize$ 
8:   for  $j = 1$  to  $m$  do
9:      $D_j \leftarrow \{(x_i, y_i) \in D \mid y_i = \omega_j\}$ 
10:     $CurrentClassSize \leftarrow |D_j|$ 
11:    if  $CurrentClassSize > NewMajoritySize$  then
12:       $index \leftarrow NewMajoritySize$  unique random integer values between 1 and  $CurrentClassSize$ 
13:       $D_{UnderSampling} \leftarrow \{(x_i, y_i) \in D_j \mid i \in index\}$ 
14:       $D_{new} \leftarrow D_{new} \cup D_{UnderSampling}$ 
15:    else
16:      Take a random sample  $D_{RandomSampling}$  of size  $CurrentClassSize$  from  $D_j$  with replacement
17:       $D_{new} \leftarrow D_{new} \cup D_{RandomSampling}$ 
18:    end if
19:  end for
20:   $D_{tr} \leftarrow$  Generate synthetic examples using  $D_{new}$ 
21:  Create
22:   $h_t \leftarrow \mathcal{L}(D_{tr})$ 
23:   $\mathcal{H} \leftarrow \mathcal{H} \cup h_t$ 
24: end for

```

In order to select the appropriate classifiers with more power in classifying minority classes examples, we propose a method that weighs the influence of each example, x_i , in the region of competence according to the proportion of examples with the same class in the neighborhood of the query example, x_t . That is, the candidate classifiers that correctly classify more minority class examples belonging to the region of competence are associated with higher competence.

Thus, we aim to describe the process of selecting an appropriate ensemble for each query example in the class imbalance scenario. The detailed procedure is described in the pseudocode in [Algorithm 2](#). Here, the key component is to assess the performance of candidate classifiers in the region of competence for each query example needed to be classified. The k nearest neighbors surrounding the query example are used to define the region of competence (lines 3–5). Our intention is to develop a DES classifier system to handle the issues present in multi-class imbalanced datasets. Hence, we aim to select the classifiers that are more powerful when classifying the examples underrepresented in the region of competence.

Algorithm 2 Pseudocode for the procedure of the dynamic selection.

```

1: Input: Training dataset  $D_{tr}$ , testing dataset  $D_{te}$ , validation dataset  $D_{va}$ , candidate classifier pool  $\mathcal{H}$ , the number of nearest neighbors  $k$ , the percentage of classifiers to be selected  $P\%$ , the scaling coefficient  $\alpha$ 
2: Output:  $EoC_t^*$ , an ensemble of classifiers for each example in  $D_{te}$ 
3: for each testing example  $x_t$  in  $D_{te}$  do
4:    $EoC_t^* \leftarrow \emptyset$ 
5:   find  $\Psi$  as the  $k$  nearest neighbors of the example  $x_t$  in  $D_{va}$ 
6:   for each example  $x_i$  in  $\Psi$  do
7:      $num \leftarrow$  count the number of examples with the same class as  $x_i$ 
8:      $w_i \leftarrow \frac{1}{1 + \exp(\alpha \times num)}$  //calculate the voting weights for each  $x_i$  in  $\Psi$ 
9:   end for
10:  Normalize  $w_i$  according to  $\widehat{w}_i \leftarrow \frac{w_i}{\sum_{i=1}^k w_i}$ 
11:  for each classifier  $h_j$  in  $\mathcal{H}$  do
12:     $C(h_j|x_t) \leftarrow \sum_{i=1}^k I(h_j(x_t) = y_t) \times \widehat{w}_i$ 
13:  end for
14:  select  $P\%$  most competent classifiers in  $\mathcal{H}$  to compose the ensemble  $EoC_t^*$  for example  $x_t$ 
15:   $H(x_t) \leftarrow \arg \max_{y \in \Omega} \sum_{i=1}^N I(h_i(x_t) = y)$ 
16: end for

```

This considered, we have developed a weighting mechanism that is described in lines 6–10, where examples in the region of competence are weighted according to their distribution. In this way, underrepresented examples in the region of competence are associated with higher weights. Then, the competence of each classifier is calculated in lines 11–13, where an adaptive weight adjustment procedure is embedded in the DES criteria. That is, classifiers with more power in classifying the difficult-to-learn examples from all the classes are associated with the higher competence. Finally, the selected classifiers are combined by majority voting rule.

4.2.3. On the computational complexity of DES-MI

Because our proposal requires to balance the training dataset using the SMOTE procedure and acquire the class distribution of the region of competence, we acknowledge that our DES-MI might be more complexity expensive than classic DES methods.

Let's assume and recall that m is the number of classes, n is the size of datasets and T is the number of base learners were already defined along the explanation. In the case of data preprocessing, we need to resample the training dataset using a random integer value for each class, where the time complexity is $\mathcal{O}(m)$. Then, the SMOTE procedure is used to balance the data distribution, in which the time complexity is $\mathcal{O}(f \cdot n_i^2)$ when calculating the Euclidean distance of any two examples for each minority class and $\mathcal{O}(n_i \log(n_i))$ for sorting the Euclidean distance in ascending order, where f is the dimension of the feature space and n_i is the number of minority class with the class label ω_i . Because we need to generate T base classifiers and n_i is significantly smaller than n , thus, we can simply the time complexity for Algorithm 1 to $\mathcal{O}(T \cdot f \cdot n^2)$.

For the procedure of the dynamic selection, the nearest neighbors of the tested example are needed to be calculated. Thus, the time complexity is $\mathcal{O}(f \cdot n_{va})$ when calculating the Euclidean distance, where n_{va} is the number of examples in the validation dataset. Then the time complexity is $\mathcal{O}(n_{va} \log n_{va})$ for sorting all current Euclidean distance calculations in ascending order. The step to find the k nearest neighbors should be $\mathcal{O}(k)$. In our scenario, f and k are both significantly smaller than n_{va} and we use the initial dataset as the validation dataset, which simplifies the time complexity of Algorithm 2 to approximately $\mathcal{O}(n \log n)$.

Therefore, we can summarize that the time complexity of the proposed DES-MI is $\mathcal{O}(n \log n + T \cdot f \cdot n^2)$ without regard to the base learner used. One should note that the computation of time complexity for our DES-MI in an empirical way is not suitable due to the strong dependence with the base learner used.

5. Experimental framework

In this section, we present the set-up of the experimental framework used to develop the empirical study in Section 6. The base learner and the parameter settings are described in Section 5.1. The details of the imbalanced datasets from real-world applications used to carry out the comparison are provided in Section 5.2. The measure metrics for evaluating the performance of the studied methods are explained in Section 5.3. Finally, the statistical tests applied to properly compare the results are presented in Section 5.4.

5.1. Base learner and parameter settings

For the generation of the candidate classifiers, the same base learner used in all the ensembles should be considered. In this study, we use classification and regression tree (CART) [3] as the baseline classifier, since CART is considered to be one of the top-10 data mining algorithms [44] and it was employed as the base learner to build an ensemble in many previous papers [23]. In CART, pruning is used and impure nodes must have 10 or more examples to be split. In the proposed DES-MI method, the candidate ensemble consists of 100 pruned decision tree classifiers ($T = 100$ in Algorithm 1) and the percentage of classifiers to be selected is set to 40% ($P = 40$ in Algorithm 2) for the dynamic selection phase. The parameter k is set to $10 \cdot m$ and the scaling coefficient α used in Algorithm 2 is set to 0.9, which was chosen using a cross-validation procedure. With respect to the approaches selected to carry out the comparison, we use the same parameter settings as those in the literature.

5.2. Datasets

In this study, twenty multi-class imbalanced datasets from the KEEL repository [36] were selected to analyze the algorithms, which are available on the corresponding web-page.¹ The description of the selected datasets is as follows:

Automobile. This dataset consists of 159 examples and each example is represented by 25 attributes and the task of this dataset is to classify a given sample as one of six types of the degree scale to which the auto is more risky than its price indicates.

Balance scale. This dataset is generated to model psychological experimental results with three types of the balance scale (tip to the right, tip to the left, and balance). Each example is represented by four attributes, i.e. the left weight, the left distance, the right weight, and the right distance.

¹ <http://www.keel.es/>

Car evaluation. This dataset is originally derived from the hierarchical decision model. It includes 1728 examples and each example is presented by six input attributes (buying, maint, doors, persons, lug_boot, safety) and one of four classes (unacceptable, acceptable, good and very good).

Heart disease (Cleveland). This is heart disease detection dataset obtained from the V.A. Medical Center. It is a five-class classification problem and each example is represented by 14 attributes and the task is to detect the presence of heart disease in the patient from zero (no presence) to four.

Contraceptive method choice. This dataset is obtained from the national Indonesia contraceptive prevalence survey. It is used to predict the current contraceptive method choice of a woman based on her demographic and socio-economic characteristics. Each example is presented by nine attributes and the contraceptive method (no use, long-term, or short-term).

Dermatology. This dataset is from a real problem in dermatology. Each example is presented by 34 attributes. It is used to classify a given sample as one of six types of dermatology (psoriasis, seboric dermatitis, lichen planus, pityriasis rosea, cronic dermatitis and pityriasis rubra pilaris).

Ecoli. In this dataset each example is represented by seven attributes and labeled by one of eight classes. The objective of this problem is to predict the localization site of proteins by employing some measures about the cell (cytoplasm, inner membrane, periplasm, outer membrane, outer membrane lipoprotein, inner membrane lipoprotein inner membrane, cleavable signal sequence).

Flare. This is a six-class classification problem with 1066 examples, in which each example is represented by eleven attributes. The objective of this problem is to predict a given sample as one of six class labels.

Glass identification. This dataset is from USA Forensic Science Service. Each example is presented by nine attributes. The task is to predict a given example as one of six types of glass which can be found in the crime scene, defined in terms of their oxide content (i.e. Na, Fe, K, etc.).

Hayes-Roth. This is an artificial dataset created to test the behavior of prototype-based classifiers. Each example is represented by four attributes (i.e. hobby, age, educational level and marital status). The hobby attribute was generated at random, thus it is employed to add noise to the data.

LED display domain. This dataset contains seven boolean attributes, one for each light-emitting diode of a seven-segment display. The task is to determine which digit is been shown in the display.

Lymphography. This dataset is a four-class classification problem obtained from the Institute of Oncology. Each example is represented by eighteen attributes and the corresponding class label.

New-thyroid. This dataset includes 215 examples with five attributes and three classes. The task is to detect a given patient is normal, suffers from hyperthyroidism, or hypothyroidism.

Page blocks classification. This dataset contains blocks of the page layout of a document that has been detected by a segmentation process. The task is to determine the type of block: Text, Horizontal line, Graphic, Vertical line or Picture.

Shuttle. Each example in this dataset is represented by nine attributes all of which are numerical. The task of this dataset is to decide what type of control of the vessel should be employed from seven possible classes (i.e. Rad Flow, Fpv Close, Fpv Open, High, Bypass, Bpv Close, and Bpv Open).

Thyroid disease. This dataset includes 720 examples from the results of Thyroid disease diagnosis and each example in this dataset is represented by 21 attributes. The task of this dataset is to detect a given patient one of three classes, i.e. normal, suffers from hyperthyroidism, or hypothyroidism.

Wine. This dataset includes the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. The task of this dataset is to classify the types of wine according to the quantities of thirteen constituents.

Red wine quality. This dataset is created using red wine samples. Each example is represented by eleven input attributes and the output is the sensory quality evaluated by wine experts. The task of this dataset is to grade the red wine quality.

Yeast. The task of this dataset is to predict the localization site of protein. Each example is represented by eight attributes and labeled with the localization site of each cell among ten possible alternatives.

Zoo. The task of this dataset is to predict the type of animal from seven predefined classes. Each example is represented by sixteen attributes and most of the attributes are boolean-valued.

The properties of the datasets are shown in Table 1, where the number of examples (#Ex.), the number of attributes (#Atts.), the number of numerical (#Num) and nominal (#Nom) attributes, the number of classes (#Cl.), the distribution of class (#Dc) and the imbalance ratio (IR) values (proportion of majority examples with respect to minority ones) are described for each dataset. The examples with missing values in the datasets (Automobile, Cleveland and Dermatology) were removed before the partitions were made.

The results were obtained by means of 3 times 5-fold stratified cross-validation (SCV) [29]. That is, each dataset was split into 5 folds with each fold containing 20% of the instances of the dataset. For each fold, the algorithm was trained with the instances contained in the remaining folds (80% of the instances of the dataset) and then tested by the current fold. The reason why a 5-fold SCV is more appropriate than a 10-fold SCV in the imbalance scenario was explained in [27]. In this paper, we use the same data partitions as in the KEEL-dataset repository [29], so that the experimental study can be reproduced by any interested reader. In this paper, we consider the training dataset as the validation dataset.

Table 1

Summary description of the datasets used in the experiments.

ID	Dataset	#Ex.	#Atts.	#Num.	#Nom.	#Cl.	#Dc.	IR
Aut	Automobile	159	25	15	10	6	46/29/13/48/20/3	16.00
Bal	Balance scale	625	4	4	0	3	288/49/288	5.88
Car	Car evaluation	1728	6	0	6	4	384/69/1210/65	18.62
Cle	Heart disease (Cleveland)	297	13	5	8	5	160/54/35/35/13	12.31
Con	Contraceptive method choice	1473	9	6	3	3	629/333/511	1.89
Der	Dermatology	358	34	1	33	6	111/60/71/48/48/20	5.55
Eco	Ecoli	336	7	7	0	8	143/77/2/2/35/20/5/52	71.50
Fla	Flare	1066	11	0	11	6	147/211/239/95/43/331	7.70
Gla	Glass identification	214	9	9	0	6	70/76/17/29/13/9	8.44
Hay	Hayes-Roth	160	4	4	0	3	65/64/31	2.10
Led	LED display domain	500	7	0	7	10	45/37/51/57/52/52/47/57/53/49	1.54
Lym	Lymphography	148	18	3	15	4	4/61/81/2	40.50
New	New-thyroid	215	5	5	0	3	150/35/30	5.00
Pag	Page blocks classification	548	10	10	0	5	491/33/3/9/12	163.67
Shu	Shuttle	2175	9	9	0	5	1706/2/6/338/123	853.00
Thy	Thyroid disease	720	21	6	15	3	17/37/666	39.18
Win	Wine	178	13	13	0	3	59/71/48	1.48
Wqr	Red wine quality	1599	11	11	0	6	10/53/681/638/199/18	68.10
Yea	Yeast	1484	8	8	0	10	463/5/35/44/51/163/244/429/20/30	92.60
Zoo	Zoo	101	16	0	16	7	41/20/5/13/4/8/10	10.25

Table 2

Confusion matrix for a binary problem.

	Predicted positive class	Predicted negative class
Actual positive class	True positive (TP)	False negatives (FN)
Actual negative class	False positives (FP)	True negatives (TN)

5.3. Performance measures

There are a large amount of metrics available for evaluating the performance of algorithms in the imbalanced datasets, e.g. precision, sensitivity, G-mean, F-measure, or AUC. However, all of them assess the classifier performing in two-class imbalanced problems. On the other hand, it would be unreasonable to use standard metrics such as classification accuracy rate in multiclass imbalanced problems, as they do not differentiate the classification rates from the different classes. Therefore, there are some proposals for measures displaying a balanced performance on multiple classes [32]. In this study, we have considered MAVa and MFM as the performance measure metrics.

The MAVa gives the same weight to each class. It achieves the accuracy rate of each class independently, and then the final result is obtained by the average value. Therefore, the MAVa is defined as the arithmetic average of the partial accuracies of each class:

$$MAVa = \frac{\sum_{i=1}^N ACC_i}{m} \quad (1)$$

where m is the number of classes and ACC_i stands for the accuracy rate for the class i .

For binary class problems, the confusion matrix (shown in Table 2) denotes results of correctly and incorrectly categorized instances. Here, the minority class is considered to be positive and the majority class is considered negative. In the two-class scenario, several measures can be deduced from the confusion matrix:

- True positive rate:

$$TP_{rate} = \frac{TP}{TP + FN} \quad (2)$$

- True negative rate:

$$TN_{rate} = \frac{TN}{FP + TN} \quad (3)$$

- False positive rate:

$$FP_{rate} = \frac{FP}{FP + TN} \quad (4)$$

- False negative rate:

$$FN_{rate} = \frac{FN}{TP + FN} \quad (5)$$

where TP_{rate} is the percentage of positive instances correctly classified as positive class, TN_{rate} is the percentage of negative instances correctly classified as positive class, FP_{rate} is the percentage of negative instances incorrectly classified as positive class and FN_{rate} is the percentage of positive instances incorrectly classified as negative class.

Recall or Sensitivity is the percentage of positive cases correctly predicted as positive class. Therefore, Recall is defined as follows:

$$Recall = TP_{rate} = \frac{TP}{TP + FN} \quad (6)$$

Conversely, Precision or Confidence is the percentage of predicted positive instances that are correctly positive, which is defined as follows:

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

F-measure is suggested in the literature to integrate precision and recall as an average, which assigns equal importance for Recall and Precision. Therefore, F-measure is defined as follows:

$$F - measure = \frac{2 \cdot recall \cdot precision}{recall + precision} \quad (8)$$

The F-measure for two-class learning problems can be modified to accommodate multi-class imbalance scenarios. MFM is defined for multi-class problems as follows:

$$MFM = \frac{\sum_{i=1}^m F - measure_i}{m} \quad (9)$$

where, i is the index of the class considered positive.

5.4. Statistical tests

In order to observe whether significant differences exist among the studied methods, it is necessary to use statistical techniques to analyze the results. Therefore, we adopt non-parametric hypothesis testing techniques recommended in the literature [7] where a set of robust and simple non-parametric tests for statistical comparison of classification algorithms is presented. We have specifically employed the Friedman aligned-ranks test [18] as a non-parametric statistical test to perform multiple comparisons among the different values of k used in our DES-MI (Section 6.1) in order to check whether significant differences among them exist or not. The Wilcoxon signed-rank test [39] is used as a statistical procedure to perform the pairwise comparison.

6. Results and discussion

In this section, we develop an extensive empirical analysis for our method with a large number of real-world imbalanced applications. We aim to verify the usefulness and effectiveness of the proposed DES-MI in the scenario of multi-class imbalanced datasets. Therefore, we first carry out the analysis of the influence of the value of k in our method in Section 6.1. Then, we present the analysis to show the effectiveness of the proposed dynamic procedure for multi-class imbalanced datasets in Section 6.2. Finally, the comparison of our method with state-of-the-art approaches is presented in Section 6.3.

6.1. Analyzing the influence of the number of nearest neighbors used in DES-MI

In this section we aim to investigate how the different configurations of k affect the performance of the proposed DES-MI. The complete results of the influence of the value of k nearest neighbors used in DES-MI are shown in Tables 3 and 4, with MAvA and MFM performance measures, respectively. The best result within each dataset is highlighted in bold-face. We have considered several configurations for $k = 1, 3, 5, 7, 9, 12, 15$ and 18 nearest neighbors. In addition, we also consider another configuration of all the instances in the validation dataset used to assess the competence of candidate classifiers, which is denoted as “all” in the last column of both tables. One should note that the proposed DES-MI still dynamically selects the most appropriate classifiers for each query example when using the configuration of “all”.

Looking at Table 3 (MAvA) and Table 4 (MFM), there are no differences found among the configurations of the k values studied when the number of datasets obtaining the best results are considered. Observing the average performance, the proposed method with the value of $k = 9$ performs better than the other values. However, we can not extract any meaningful conclusions without the proper statistical analysis. Hence, in order to detect whether differences exist among the values of k studied, we will execute the non-parametric statistical tests for the comparison.

The Friedman aligned-rank test is employed in this study to analyze whether significant differences are present when using different configurations for setting the k value. The results of the statistical analysis do not reject the null hypothesis that all the configurations are equivalent, since the p -values returned by the Friedman aligned-rank test are higher than our α -value (0.05) for both performance measures, MAvA (0.516020) and MFM (0.081551). On account of these experimental results, we set the value of $k = 9$ for the rest of this paper, as it achieves the best average performance, for both MAvA and kappa performance measures.

Table 3

Average MAVa test results for different values of k in DES-MI (“all” means all the instances in the validation dataset used to assess the competence of candidate classifiers). The best result in each dataset is highlighted in bold-face.

ID	$k = 1$	$k = 3$	$k = 5$	$k = 7$	$k = 9$	$k = 12$	$k = 15$	$k = 18$	$k = all$
Aut	85.69	84.23	82.44	84.01	86.30	83.88	84.11	83.86	83.65
Bal	57.60	59.65	59.62	62.26	63.89	61.53	62.83	61.36	62.90
Car	94.09	93.40	93.27	93.94	93.56	94.16	94.54	95.12	96.42
Cle	31.05	29.55	30.33	29.63	30.55	30.75	31.37	30.71	30.90
Con	50.55	50.62	50.21	50.80	50.64	51.47	51.05	50.96	53.29
Der	96.15	96.27	96.20	96.45	96.00	96.01	95.30	95.30	95.39
Eco	72.35	73.06	74.45	74.08	72.69	72.89	72.41	72.44	73.53
Fla	65.41	63.92	64.55	65.48	65.21	64.81	64.56	64.49	64.54
Gla	76.63	76.76	75.73	76.70	76.48	77.01	76.73	76.17	77.47
Hay	89.36	88.16	89.06	86.82	88.16	86.44	86.62	87.82	85.26
Led	71.43	70.44	71.55	72.71	73.32	73.22	72.83	72.50	71.88
Lym	82.98	83.51	84.05	84.39	84.74	84.17	83.96	82.99	83.79
New	95.12	95.62	94.51	93.87	93.87	94.56	94.47	94.39	93.59
Pag	72.75	79.82	79.32	77.77	77.76	77.91	79.58	79.55	75.02
Shu	91.80	91.80	91.80	91.80	91.80	91.79	91.79	91.79	93.11
Thy	91.15	91.20	91.94	91.94	91.94	92.68	92.68	92.66	97.46
Win	95.46	95.80	95.62	96.03	96.18	95.77	95.22	94.91	95.85
Wqr	41.56	41.11	40.68	43.21	42.89	41.75	41.24	41.51	42.36
Yea	56.83	57.07	57.85	58.26	58.26	57.66	58.05	57.40	57.89
Zoo	88.57	90.95	91.43	91.31	91.79	90.83	90.83	90.83	90.83
Avg.	75.33	75.65	75.73	76.07	76.30	75.96	76.01	75.84	76.26

Table 4

Average MFM test results for different values of k in DES-MI (“all” means all the instances in the validation dataset used to assess the competence of candidate classifiers). The best result in each dataset is highlighted in bold-face.

ID	$k = 1$	$k = 3$	$k = 5$	$k = 7$	$k = 9$	$k = 12$	$k = 15$	$k = 18$	$k = all$
Aut	0.8366	0.8178	0.7932	0.8216	0.8438	0.8182	0.8171	0.8110	0.8083
Bal	0.5790	0.6036	0.6005	0.6150	0.6274	0.6106	0.6179	0.6101	0.6278
Car	0.9434	0.9430	0.9413	0.9472	0.9422	0.9471	0.9486	0.9488	0.9420
Cle	0.3009	0.2866	0.2934	0.2875	0.2945	0.2941	0.2978	0.2946	0.2997
Con	0.5044	0.5044	0.4996	0.5043	0.5027	0.5084	0.5053	0.5048	0.5254
Der	0.9585	0.9601	0.9584	0.9620	0.9568	0.9570	0.9469	0.9457	0.9456
Eco	0.6585	0.6613	0.6766	0.6724	0.6574	0.6549	0.6541	0.6516	0.6688
Fla	0.6367	0.6175	0.6205	0.6267	0.6245	0.6203	0.6216	0.6220	0.6232
Gla	0.7471	0.7599	0.7335	0.7379	0.7401	0.7434	0.7467	0.7413	0.7538
Hay	0.8928	0.8800	0.8897	0.8662	0.8798	0.8623	0.8640	0.8763	0.8500
Led	0.7086	0.6965	0.7063	0.7212	0.7284	0.7271	0.7224	0.7191	0.7106
Lym	0.7722	0.7861	0.7915	0.7955	0.7904	0.7743	0.7666	0.7526	0.7643
New	0.9569	0.9615	0.9460	0.9419	0.9419	0.9463	0.9431	0.9347	0.9237
Pag	0.6973	0.7649	0.7484	0.7321	0.7300	0.7332	0.7424	0.7415	0.7125
Shu	0.9074	0.9074	0.9074	0.9074	0.9074	0.8962	0.8962	0.8962	0.8883
Thy	0.9028	0.9070	0.9115	0.9115	0.9115	0.9160	0.9160	0.9150	0.9345
Win	0.9510	0.9547	0.9529	0.9585	0.9606	0.9566	0.9511	0.9479	0.9564
Wqr	0.4032	0.3969	0.3880	0.4022	0.4011	0.3946	0.3889	0.3876	0.3951
Yea	0.5488	0.5460	0.5556	0.5553	0.5550	0.5505	0.5571	0.5514	0.5613
Zoo	0.8615	0.8872	0.8932	0.8902	0.8961	0.8847	0.8847	0.8847	0.8847
Avg.	0.7384	0.7421	0.7404	0.7428	0.7446	0.7398	0.7394	0.7368	0.7388

6.2. Analyzing the effectiveness of the proposed dynamic procedure for multi-class imbalanced datasets

In this section, we aim to answer the question “Is it effective to use the proposed DES procedure to improve the classification performance in the scenario of multi-class imbalanced datasets?”. This considered, we carry out the experimental analysis to compare our proposed DES-MI with the representative benchmark approaches, i.e. the initial base learner, SMOTE, dynamic selection of a single classifier, and combination of all the candidate classifiers. The description of the representative approaches is as follows:

- The base classifier (denoted as Base): This method builds the classifier using the base classification algorithm on the original training dataset.
- Resampling the dataset using SMOTE (denoted as SMOTE): In this method, the training dataset is balanced by the usage of the SMOTE algorithm. Then, the classifier is trained on the balanced dataset by using the base learner.

Table 5

Average MAvA and MFM test results of the representative approaches and our DES-MI. The best result in each dataset for each performance measure is highlighted in bold-face.

ID	MAvA					MFM				
	Base	SMOTE	DCS-MI	MV	DES-MI	Base	SMOTE	DCS-MI	MV	DES-MI
Aut	72.76	76.40	82.42	84.59	86.30	0.6948	0.7336	0.8007	0.8257	0.8438
Bal	57.52	58.53	58.03	61.54	63.89	0.5595	0.5899	0.5883	0.6099	0.6274
Car	88.43	90.82	93.55	96.67	93.56	0.8810	0.9165	0.9412	0.9506	0.9422
Cle	29.00	29.10	30.75	29.20	30.55	0.2875	0.2804	0.2947	0.2849	0.2945
Con	48.98	48.53	47.24	53.98	50.64	0.4899	0.4846	0.4710	0.5360	0.5027
Der	91.75	94.38	93.45	96.20	96.00	0.9225	0.9406	0.9306	0.9559	0.9568
Eco	61.93	69.58	70.69	73.83	72.69	0.5578	0.6117	0.6305	0.6726	0.6574
Fla	61.11	63.99	62.75	64.86	65.21	0.6157	0.6234	0.6260	0.6308	0.6245
Gla	65.10	68.94	74.22	75.74	76.48	0.6590	0.6800	0.7233	0.7359	0.7401
Hay	85.60	85.43	84.77	84.40	88.16	0.8546	0.8528	0.8464	0.8406	0.8798
Led	71.01	71.31	69.61	72.62	73.32	0.7044	0.7079	0.6883	0.7196	0.7284
Lym	68.03	83.25	84.18	84.29	84.74	0.6023	0.7736	0.7858	0.7894	0.7904
New	88.35	91.68	90.66	93.07	93.87	0.8809	0.9121	0.9120	0.9394	0.9419
Pag	81.19	69.09	73.65	75.85	77.76	0.7060	0.6612	0.6928	0.7310	0.7300
Shu	75.77	91.83	93.18	91.79	91.80	0.7267	0.9071	0.9190	0.8906	0.9074
Thy	92.64	89.53	90.23	93.37	91.94	0.9272	0.8972	0.8912	0.9164	0.9115
Win	92.50	94.36	94.05	95.95	96.18	0.9223	0.9427	0.9376	0.9566	0.9606
Wqr	32.27	36.72	37.22	41.94	42.89	0.3299	0.3232	0.3534	0.4041	0.4011
Yea	39.22	52.96	50.92	59.15	58.26	0.3967	0.4970	0.4786	0.5744	0.5550
Zoo	77.14	87.14	89.88	88.10	91.79	0.7222	0.8429	0.8764	0.8518	0.8961
Avg.	69.01	72.68	73.57	75.86	76.30	0.6721	0.7089	0.7194	0.7408	0.7446

Table 6

Results of Wilcoxon tests for comparing DES-MI with Base, SMOTE, DCS-MI and MV, respectively. A “*” near the p -value means that there are statistical differences with $\alpha = 0.05$ (95% confidence).

Performance measure	Comparison	R^+	R^-	Hypothesis	p -value
MAvA	DES-MI vs. Base	203.00	7.00	Rejected for DES-MI at 5%	0.000254*
	DES-MI vs. SMOTE	209.00	1.00	Rejected for DES-MI at 5%	0.000103*
	DES-MI vs. DCS-MI	204.00	6.00	Rejected for DES-MI at 5%	0.000219*
	DES-MI vs. MV	140.00	70.00	Not rejected	0.191334
MFM	DES-MI vs. Base	206.00	4.00	Rejected for DES-MI at 5%	0.000163*
	DES-MI vs. SMOTE	210.00	0.00	Rejected for DES-MI at 5%	0.000089*
	DES-MI vs. DCS-MI	201.00	9.00	Rejected for DES-MI at 5%	0.000338*
	DES-MI vs. MV	127.00	83.00	Not rejected	0.411465

- Dynamic classifier selection for multi-class imbalanced datasets (denoted as DCS-MI): The generation of candidate classifiers in this method is the same as that in our proposal, but DCS-MI selects the most appropriate classifier rather than an ensemble for each query example.
- Majority Voting (denoted as MV): In this method, an ensemble is obtained by combining all the candidate classifiers with majority voting.

Observing the results presented in Table 5, where the best result in each dataset for each performance measure is highlighted in bold-face, we can easily note that the proposed DES-MI outperforms the other compared methods in both performance measures. The average performance in both measures is remarkable, as is the number of datasets in which DES-MI is the best method. However, we cannot obtain meaningful conclusions without carrying out the proper statistical analysis.

Hence, we have performed the required Wilcoxon tests in order to compare the proposed method DES-MI with each method used in this section for both performance measures. The completed statistical results are shown in Table 6, where R^+ corresponds to the sum of the ranks for DES-MI and R^- for the compared method. Observing Table 6, we can find that DES-MI is significantly better than Base, SMOTE and DCS-MI, since the corresponding p -values obtained are lower than our α -value (0.05). Although there are no significant differences obtained in the comparison of DES-MI and MV for both performance measures, we can still highlight the robust performance of the proposed DES-MI, as in both cases the values of R^+ are considerably greater than that of R^- . Therefore, we can highlight the effectiveness of the proposed dynamic procedure in addressing multi-class imbalanced datasets.

6.3. Comparison of DES-MI versus state-of-the-art methods

As we have mentioned above, we in this subsection aim to execute the comparison of our DES-MI with the state-of-the-art techniques developed for multi-class imbalanced datasets. There are a number of solutions already proposed in the

Table 7

Average MAvA and MFM test results of the state-of-the-art methods and DES-MI. The best result in each dataset for each performance measure is highlighted in bold-face.

ID	MAvA					MFM				
	OVO-Easy	OVA-NBSVM	AdaBoost.NC	Static-SMOTE	DES-MI	OVO-Easy	OVA-NBSVM	AdaBoost.NC	Static-SMOTE	DES-MI
Aut	73.84	68.85	69.73	74.80	86.30	0.6696	0.6728	0.6245	0.7309	0.8438
Bal	68.58	73.09	62.86	57.87	63.89	0.6300	0.7387	0.6098	0.5774	0.6274
Car	97.65	91.64	96.09	94.07	93.56	0.9350	0.9330	0.9481	0.9302	0.9422
Cle	31.99	28.46	31.60	28.19	30.55	0.2920	0.2743	0.3027	0.2726	0.2945
Con	50.85	45.13	53.53	47.01	50.64	0.5014	0.3777	0.5221	0.4701	0.5027
Der	93.20	92.97	95.55	92.70	96.00	0.9264	0.9407	0.9526	0.9236	0.9568
Eco	57.09	69.35	68.19	65.80	72.69	0.5244	0.6337	0.5958	0.6029	0.6574
Fla	63.70	58.94	59.23	62.62	65.21	0.5998	0.5899	0.5591	0.6050	0.6245
Gla	72.43	51.62	71.57	70.79	76.48	0.6537	0.5108	0.6286	0.6855	0.7401
Hay	85.46	61.97	69.94	84.74	88.16	0.8531	0.5628	0.6634	0.8451	0.8798
Led	70.59	66.32	42.34	70.87	73.32	0.6979	0.6477	0.3674	0.7045	0.7284
Lym	84.53	61.44	88.01	78.41	84.74	0.6942	0.5338	0.7907	0.7163	0.7904
New	95.88	74.29	90.79	90.62	93.87	0.9222	0.8069	0.8970	0.9006	0.9419
Pag	80.82	56.08	73.42	81.71	77.76	0.6185	0.5282	0.6875	0.7632	0.7300
Shu	87.31	54.80	91.78	90.26	91.80	0.7699	0.4910	0.8714	0.8777	0.9074
Thy	96.85	42.71	97.31	92.36	91.94	0.8494	0.4455	0.9374	0.9257	0.9115
Win	95.24	98.34	89.24	93.02	96.18	0.9496	0.9825	0.8894	0.9289	0.9606
Wqr	40.53	32.71	39.44	33.33	42.89	0.2849	0.2963	0.3079	0.3268	0.4011
Yea	52.63	46.21	54.14	50.08	58.26	0.4691	0.4705	0.4609	0.4930	0.5550
Zoo	85.95	87.14	87.74	89.05	91.79	0.8230	0.8443	0.8427	0.8597	0.8961
Avg.	74.26	63.10	71.63	72.41	76.30	0.6832	0.6140	0.6729	0.7070	0.7446

Table 8

Results of Wilcoxon test for the pairwise comparison between DES-MI and the state-of-the-art methods for both performance measures (MAvA and MFM). A “*” near the p -value means that there are statistical differences with $\alpha = 0.05$ (95% confidence).

Performance measure	Comparison	R^+	R^-	Hypothesis	p -value
MAvA	DES-MI vs. OVO-Easy	143.00	67.00	Not rejected	0.156004
	DES-MI vs. NBSVM	197.00	13.00	Rejected for DES-MI at 5%	0.000593*
	DES-MI vs. AdaBoost.NC	172.00	38.00	Rejected for DES-MI at 5%	0.012374*
	DES-MI vs. Static-SMOTE	194.00	16.00	Rejected for DES-MI at 5%	0.000892*
MFM	DES-MI vs. OVO-Easy	207.00	3.00	Rejected for DES-MI at 5%	0.000140*
	DES-MI vs. NBSVM	195.00	15.00	Rejected for DES-MI at 5%	0.000780*
	DES-MI vs. AdaBoost.NC	189.00	21.00	Rejected for DES-MI at 5%	0.001713*
	DES-MI vs. Static-SMOTE	198.00	12.00	Rejected for DES-MI at 5%	0.000517*

literature for handling skewed distributions in classes. We have selected four well-known techniques from the literature as reference methods for our proposal, i.e. OVO-Easy [48], OVA-NBSVM [6], AdaBoost.NC [37] and Static-SMOTE [11].

From Table 7, one can observe that the proposed DES-MI performs considerably better than the selected state-of-the-art techniques, both in terms of MAvA and MFM performance measures. Concretely, DES-MI achieves the best results on 11 out of 20 datasets when MAvA is used as the performance measure, whereas with MFM as the performance measure DES-MI performs best on 12 out of 20 datasets. With respect to the average performance, DES-MI also outperforms the selected state-of-the-art methods.

Again, one can not extract any meaningful conclusions without the appropriate statistical analysis. Therefore, we have followed the same methodology used in the previous section. We have also employed the Wilcoxon test to compare our DES-MI with the state-of-the-art techniques for both performance measures. The results of these tests are shown in Table 8, where R^+ corresponds to the sum of the ranks for DES-MI and R^- for the out-of-the-state method.

From Table 8, we can observe that all the null hypotheses are rejected for DES-MI except the comparison between DES-MI and OVO-Easy. With respect to the pairwise comparison between DES-MI and OVO-Easy, one can observe the significant difference between them when MFM is used as the performance measure, whereas considering MAvA the null hypothesis of equivalence is not rejected with the p -value of 0.156004. However, the value of R^+ (143.00) corresponding to the sum of the ranks for DES-MI is obviously greater than R^- (67.00) for OVO-Easy and the obtained p -value is low, which indicates the superiority of the DES-MI with respect to OVO-Easy. Therefore, we can highlight the goodness of the proposed method using DES-based frameworks for multi-class imbalanced datasets.

7. Conclusions

In this paper, we have proposed a novel DES-based procedure to deal with multi-class imbalanced datasets, called DES-MI. The proposed DES-MI method consists of two key components: the generation of a balanced training dataset, in which a novel data preprocessing method is proposed to generate a balanced training dataset, and the selection of appropriate classifiers, in which the competence of candidate classifiers is evaluated using the weighted region of competence and in which classifiers with more power in classifying the minority class instances have more possibility to be selected to construct the ensemble for each query example.

The proposed DES-MI method aims to improve the classification performance in the scenario of multi-class imbalance problems by selecting an appropriate subset of classifiers for each query example. Experiments were conducted using twenty multi-class imbalanced datasets. The analysis of the proposed selection procedure indicates its effectiveness in dealing with multi-class imbalanced datasets. In addition, the performance obtained by the proposed DES-MI is compared with four state-of-the-art techniques devoted to handling the multi-class imbalance learning problems and experimental results demonstrated that the proposed DES-MI outperforms the previous techniques in the majority of datasets.

The proposed method provides a direction for the future research to extend the strategies proposed for constructing a DES classifier system in a multi-class imbalance scenario. In the future, there are several works remaining to be addressed. Among them, the scalability of DES-based method and more specifically, of our DES-MI must be studied. Furthermore, it would be interesting to extend our technique to semi-supervised learning with unseen labels [35].

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